



Bayesian analysis of repairable systems with modulated power law process



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ABSTRACT

As a compromise between nonhomogeneous Poisson process and renewal process, the modulated power law process is more appropriate to model the failures of repairable systems. In this article, objective Bayesian methods are proposed to analyze the modulated power law process. Seven reference priors, one of which is also the Jeffreys prior, are derived. However, only four of them are taken into consideration because of their practicality. Propriety of the posterior densities considering the four reference priors is proved. Predictive distribution of the future failure time is obtained additionally. For the purpose of comparison, the simulation work and real data analysis are carried out based on both the objective Bayesian and maximum likelihood approaches, which show that the objective Bayesian estimation and prediction have much better statistical properties in a frequentist context, and outperforms the maximum likelihood method even with small or moderate sample sizes.

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1. Introduction

In reliability theory, much attention has been paid to non-repairable system (unit, product) that can fail only once. The observation of a non-repairable system (unit, product) is known as the lifetime which may often be modeled by the exponential, Weibull or gamma distributions. However, most of the systems can be repaired, and the failure of a repairable system occurs repeatedly during a given period. The observations consist of a series of repair times or repair costs. Such an event is termed as a *recurrent event* in other fields like clinical study and software reliability. For more details, see Gail et al. [1], Dalal and McIntosh [2], and Therneau and Hamilton [3].

Stochastic point processes have been widely used to analyze lifetime data of repairable systems since they can characterize the failures by assuming they are distributed randomly in time. Nonhomogeneous Poisson process (NHPP) and renewal process (RP) are probably two of the most commonly used ones to describe the failure pattern. In some cases, the repairable system returns to the same state as before the failure after a repair. In other cases, the system becomes a new one after a repair. NHPP can depict the effectiveness of repairable system achieved by “same-as old” (minimal repair assumption) and RP can depict the effectiveness achieved by “same-as new” (perfect maintenance assumption). That is, both NHPP and RP perform well in the ideal world and are easy to handle. Nevertheless, in practice both “same-as old” and “same-as new” are two extreme failure pattern of repairable system, since repair can make a system perform better

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than it was before the failure occurrence but worse than a brand-new one. The NHPP seems pessimistic whereas RP is overly optimistic to fit the problem of repairable system. In public health, in addition, some diseases occur again and again after treatment and these two processes are obviously too simple to describe the machinery of life. Accordingly, models applied for analyzing recurrent event data must be capable of simultaneously delineating the relationship between current event and previous treatment and telling whether the individual experiences deterioration or improvement as it ages. These models must be inherently different from those for lifetime data. There are some scenarios that the NHPP model can be correctly used in reliability. For example, the system is complex and repair consists only in substituting the failed part with a new one which does little good or harm to anything else in the system. A complete discussion on repairable system and recurrent event data can be found in [4] and [5]. In this paper, we concentrate on discussing the recurrent event data from reliability, i.e. observations from repairable system, and assume that repair is provided as soon as the system fails.

To give a compromise between the nonhomogeneous Poisson process and renewal process, Lakey and Rigdon [6] introduced the modulated power law process (MPLP). It is an extension of NHPP with power or Weibull intensity which is often referred to as power law process (PLP). and also a special case of the inhomogeneous Gamma process defined by Berman [7]. For the modulated power law process, the failure probability at a given time t of the repairable system depends not only on the age of the system but also on the recent repair time. Thus, it can incorporate both renewal-type behavior and time trend. The MPLP is more suitable to model the repair or failure patterns in many practical settings in reliability. The MPLP can be used to analyze a wide range of point processes, such as the failure process of repairable systems, the duration dependence of business cycles [8], and stock market cycles [9]. Approaches used for exploring the MPLP so far are maximum likelihood method and subjective Bayesian method. Black and Rigdon [10] proposed an algorithm for deriving the maximum likelihood estimates for the model parameters. Asymptotic analysis served to give approximate confidence intervals and hypothesis tests for the parameters. Coverage probabilities of the asymptotic confidence intervals were strictly less than the nominal level in some settings. From the Bayesian perspective, Calabria and Pulcini [11] elicited non-informative and vague joint priors to proceed Bayesian analysis. In their paper, the non-informative prior was actually the independent Jeffreys prior which was obtained based on the independent assumption of parameters. Such assumption might be subjective and inappropriate, and make the “non-informative” prior informative. Moreover, propriety of the posteriors has not been proved in their article. Bayesian hypothesis tests for the parameters were further studied by Calabria and Pulcini [12]. Furthermore, Muralidharan [13] also presented various tests for the model parameters in the presence of nuisance parameters. Reliability of the i th process of MPLP was studied by Muralidharan [13], where the maximum likelihood estimate and the uniformly minimum variance unbiased estimate of it were derived.

Besides the two approaches mentioned above, in this article, however, we investigate the modulated power law process via an objective Bayesian approach. It is a method that has not been used to analyze such a model before. As we all know, choosing an appropriate prior is the main task for Bayesian statisticians. In practice, subjective priors always depend on the experts' belief which is neither easy to quantify in a limited time period nor easy to convince others. Therefore, given little prior information, we prefer to use objective priors to make inference. A reference prior is one of the objective priors and introduced by Bernardo [14]. Other objective priors include Jeffreys prior, probability matching priors, and so on. Different objective priors are defined under different frameworks. In our paper, only reference priors are considered. The basic idea of constructing a reference prior is to construct a density maximizing the missing information about the parameter of interest. Via the objective Bayesian approach, we succeed in doing Bayesian estimation and prediction whose coverage probabilities are close to the nominal level even for small samples. Because of the characteristics of reference priors, they could conveniently be used as standards for scientific communication.

The motivation for us to consider the MPLP from the view of objective Bayes has been mentioned previously. The paper is organized as following. In Section 2, model formulation of the modulated power law process is given, where the Fisher information matrix of the likelihood and Jeffreys prior are calculated. In Section 3, seven reference priors, one of which is also Jeffreys prior, are derived. Some of them perform poorly for their complicated forms and setting undesirable constraints to the parameters. Therefore, only four of the reference priors are taken into consideration for practicality. Bayesian posterior analysis considering the four reference priors are given in Section 4. The propriety of posterior distributions is proved. Procedures for Gibbs sampling are also given in this section. Bayesian prediction of the future failure time is investigated in Section 5. Simulation studies and real data analysis are given in Sections 6 and 7, respectively. The statistical properties of Bayesian estimates are compared, via Monte Carlo simulation, with the maximum likelihood ones in a frequentist context in the simulation studies. Both simulation and application work show the superiority of objective Bayes over classical maximum likelihood method. Therefore, the Bayesian approach using reference priors can be an attractive alternative way to analyze the MPLP.

2. Model formulation

Consider a repairable system experiencing shocks that occur according to the NHPP with power law or Weibull intensity function,

$$h(t) = \lim_{\Delta t \rightarrow 0} \frac{P[\text{a shock occurs in } (t, t + \Delta t)]}{\Delta t} = \frac{\beta}{\theta} \left(\frac{t}{\theta} \right)^{\beta-1}, \quad t \geq 0, \mathcal{L} \quad (1)$$

where $\beta > 0$ and $\theta > 0$ are the shape and scale parameters respectively. The intensity function represents the rate for a shock occurring at time t . Let

$$H(t) = \int_0^t h(u) du = \left(\frac{t}{\theta}\right)^\beta$$

be the corresponding cumulative intensity function that depicts the expectation of the number of shocks occurring in period $[0, t)$. Suppose the system fails when the κ th shock occurs, where κ is a positive integer. For instance, when $\kappa = 3$, the system breaks at every third shock. Assume that the system will be fixed as soon as it fails. Therefore, the repaired system would be in a better condition than it was in before the failure occurrence, since three other shocks must occur in order to observe the next failure.

Let $t_1 < t_2 < \dots < t_n$ be the recurrent failures or repair times of the repairable system which are failure-truncated samples. Based on the previous “shock” example, the modulated power law process (MPLP) is defined as following. For any $\kappa > 0$, $H(t_i) - H(t_{i-1})$ ($i = 1, \dots, n$), where $t_0 = 0$, are independently and identically Gamma distributed with unknown shape parameter κ and unit scale parameter. That is,

$$\left(\frac{t_i}{\theta}\right)^\beta - \left(\frac{t_{i-1}}{\theta}\right)^\beta \stackrel{i.i.d.}{\sim} \text{Gamma}(\kappa, 1), \quad i = 1, \dots, n. \quad (2)$$

In the MPLP model, $\mu = (\beta, \theta, \kappa)$ are the parameters to be estimated. Both κ and β possess clear but different physical meaning. The value of κ is a measure of the improvement or worsening of the system after the repair actions. For $\kappa > 1$, the system is in a better condition after the fixing and the larger κ is, the larger improvement will be. For $0 < \kappa < 1$, the system is in a worse condition after a repair which may rarely happen in practice. On the other hand, the value of β is a measure of improvement or deterioration over the system life. Here are some special cases for MPLP model.

- When $\kappa = 1$, the MPLP reduces to the PLP.
- When $\beta = 1$, the MPLP reduces to the Gamma renewal process.
- When $\kappa = \beta = 1$, the MPLP becomes the homogeneous Poisson process.

In this paper, we consider the scenario where κ is a positive real number. On the basis of condition (2), the conditional probability density function of t_i given t_{i-1} is

$$f(t_i|t_{i-1}) = \frac{1}{\Gamma(\kappa)} \frac{\beta}{\theta} \left(\frac{t_i}{\theta}\right)^{\beta-1} \left[\left(\frac{t_i}{\theta}\right)^\beta - \left(\frac{t_{i-1}}{\theta}\right)^\beta \right]^{\kappa-1} \times \exp \left\{ -\left(\frac{t_i}{\theta}\right)^\beta + \left(\frac{t_{i-1}}{\theta}\right)^\beta \right\}, \quad t_i > t_{i-1},$$

for $i = 1, \dots, n$. Hence, the likelihood function of the observations can be written as,

$$\begin{aligned} L(\mathbf{t}|\mu) &= f(t_1)f(t_2|t_1)\dots f(t_n|t_{n-1}) \\ &= \frac{1}{\Gamma(\kappa)^n} \beta^n \theta^{-n\beta\kappa} \prod_{i=1}^n t_i^{\beta-1} \prod_{i=1}^n (t_i^\beta - t_{i-1}^\beta)^{\kappa-1} \exp \left\{ -\left(\frac{t_n}{\theta}\right)^\beta \right\}, \end{aligned} \quad (3)$$

where $\mathbf{t} = (t_1, \dots, t_n)$ are the observations. The corresponding log-likelihood is

$$l(\mathbf{t}|\mu) = -n \log \Gamma(\kappa) + n \log \beta - n\beta\kappa \log \theta + (\beta - 1) \sum_{i=1}^n \log t_i + (\kappa - 1) \sum_{i=1}^n \log(t_i^\beta - t_{i-1}^\beta) - \left(\frac{t_n}{\theta}\right)^\beta. \quad (4)$$

If κ is positive but not an integer, the expression shown in (3) is still a joint probability density function. Thus the MPLP can be defined for any positive value of κ . We give a simple explanation of κ when κ is not an integer. Let $X_1, X_2, \dots, X_n$ be n i.i.d. gamma random variables with shape parameter κ and unit scale parameter. This distribution has mean $\mu = \kappa$ and standard deviation $\sigma = \sqrt{\kappa}$. Now define

$$\begin{aligned} Y_1 &= X_1 \\ Y_2 &= X_1 + X_2 \\ &\vdots \\ Y_n &= X_1 + X_2 + \dots + X_n. \end{aligned}$$

The Y_i 's thus represent the event times of a renewal process in which the times between events are i.i.d. gamma. Now define $T_i = \theta Y_i^{1/\beta}$, $i = 1, 2, \dots, n$. The process $T_1 < T_2 < \dots < T_n$ is then an MPLP with parameters θ , β , and κ .

In order to obtain the objective priors of the parameters, the Fisher information matrix is needed and given by,

$$I(\beta, \theta, \kappa) = \begin{pmatrix} \frac{1}{\beta^2} f_1(\kappa) & \frac{1}{\theta} f_2(\kappa) & \frac{1}{\beta} f_3(\kappa) \\ \frac{1}{\theta} f_2(\kappa) & \frac{n\kappa\beta^2}{\theta^2} & \frac{n\beta}{\theta} \\ \frac{1}{\beta} f_3(\kappa) & \frac{n\beta}{\theta} & n\psi^{(1)}(\kappa) \end{pmatrix}, \quad (5)$$

where

$$\begin{aligned} f_1(\kappa) &= n + n\kappa[\psi(1 + n\kappa)^2 + \psi^{(1)}(n\kappa)] \\ &\quad + \kappa \sum_{i=1}^n (i-1) \{[\psi(ik - \kappa + 1) - \psi(ik) + 1]^2 + \psi^{(1)}(ik - \kappa + 1) - \psi^{(1)}(ik) - 1\}, \\ f_2(\kappa) &= -n\kappa\psi(n\kappa + 1), \quad \text{and} \\ f_3(\kappa) &= -\frac{1}{\kappa - 1} \left[n\kappa\psi(1 + n\kappa) - n - \sum_{i=1}^n \psi(ik) \right], \quad \kappa \neq 1. \end{aligned}$$

$f_1(\kappa)$, $f_2(\kappa)$ and $f_3(\kappa)$ given above are continuous functions depending on κ only with

$$\psi(x) = \frac{d \log \Gamma(x)}{dx}, \quad \psi^{(m)}(x) = \frac{d^m \psi(x)}{dx^m} \quad \text{for } m \in \{1, 2, \dots\},$$

the digamma and polygamma functions, respectively.

The Fisher information matrix (5) has a fussy form. Note that the off-diagonal elements are nonzero. Thus, the assumption that parameters were a priori independent made in [11] was inappropriate and Jeffreys prior elicited in the paper was not a non-informative one as they said. The determinant of Fisher information matrix (5) is

$$|I(\beta, \theta, \kappa)| = \frac{ng(\kappa)}{\theta^2},$$

where

$$g(\kappa) = n\kappa\psi^{(1)}(\kappa)f_1(\kappa) - nf_1(\kappa) - \psi^{(1)}(\kappa)f_2(\kappa)^2 + 2f_2(\kappa)f_3(\kappa) - \kappa f_3(\kappa)^2$$

is a function of κ . Accordingly, Jeffreys' prior of $\mu = (\beta, \theta, \kappa)$ should be

$$\pi_J(\mu) \propto \frac{\sqrt{g(\kappa)}}{\theta}, \quad \theta > 0, \quad g(\kappa) > 0. \quad (6)$$

Without any priori independent assumptions about the parameters, the Jeffreys' prior has too poor performance to be used as a non-informative prior in objective Bayesian analysis. First of all, the unfriendly form of $g(\kappa)$ makes Bayesian analysis intractable. It also sets an undesirable constraint on parameter κ so that $g(\kappa) > 0$ would be satisfied. For example, with sample size $n = 15$, κ has a constraint: $0 < \kappa < 1.667$. Hence, the posterior estimate of κ will always underestimate the true value. In addition, the Jeffreys' prior (6) Does not contain the shape parameter β and it might behave worse than priors considering β . Jeffreys' prior performs great in the setting where the parameter is of one dimension but may experience difficulties when the parameters are multi-dimensional.

3. Reference priors

In objective Bayesian analysis, a reference prior is another type of non-informative priors. It was firstly introduced by Bernardo [14] and widely discussed in [15,16]. As a refinement of the Jeffreys' prior for higher dimensional problem, it is defined based on the Kullback–Leibler (K–L) divergence. By maximizing the expectation of the K–L divergence from posterior to prior distributions, known as the missing information, the reference prior can be constructed. For the meaning of “non-informative”, the maximized missing information guarantees the reference prior provides as little information as it can.

In the one dimensional case, the reference prior is equivalent to Jeffreys' prior. With model containing more than one parameter, the parameters could be partitioned into several groups according to their degree of interest to practitioners. The more interesting the parameter is, the earlier its position is. The least interesting ones are known as nuisance parameters. Switching the role of parameters of interest and nuisance ones will sometimes give a reasonable reference prior (see [17]). For certain group ordering of parameters, the reference prior is calculated step by step through maximizing the missing information on the basis of the conditional priors on some compact sets. The compact sets are constructed for the purpose of propriety.

In our model, the parameters μ are identified into 2 or 3 groups. For example, by dividing μ into 3 groups, we have $3! = 6$ patterns of the parameters which include $\{\beta, \theta, \kappa\}$, $\{\beta, \kappa, \theta\}$, $\{\theta, \beta, \kappa\}$, $\{\theta, \kappa, \beta\}$, $\{\kappa, \beta, \theta\}$ and $\{\kappa, \theta, \beta\}$. If we treat all parameters the same just as we only have one parameter, there will be only one group of them and the corresponding reference prior will be the Jeffreys' prior. For three groups of μ , we have following results.

Theorem 1. When the grouping order is $\{\beta, \theta, \kappa\}$ or $\{\theta, \beta, \kappa\}$, the reference prior of μ is

$$\pi_1(\mu) \propto \frac{1}{\beta\theta} [\psi^{(1)}(\kappa)]^{1/2}, \quad \beta, \theta, \kappa > 0. \quad (7)$$

Proof. Given the grouping order $\{\beta, \theta, \kappa\}$, we mean that β is the parameter of the most interest and θ is of less interest while κ is the nuisance one. Following the notations in [15], we derive

$$h_1 = \frac{g(\kappa)}{n\beta^2[\kappa\psi^{(1)}(\kappa) - 1]}, \quad h_2 = \frac{n\beta^2[\kappa\psi^{(1)}(\kappa) - 1]}{\theta^2\psi^{(1)}(\kappa)}, \quad h_3 = n\psi^{(1)}(\kappa),$$

which serve to maximize the missing information. Choose $\Omega_s = [\beta_{1s}, \beta_{2s}] \times [\theta_{1s}, \theta_{2s}] \times [\kappa_{1s}, \kappa_{2s}]$ as a collection of compact sets in $(0, \infty) \times (0, \infty) \times (0, \infty)$ such that $\beta_{1s}, \theta_{1s}, \kappa_{1s} \rightarrow 0$ and $\beta_{2s}, \theta_{2s}, \kappa_{2s} \rightarrow \infty$ as $s \rightarrow \infty$. The conditional prior of nuisance parameter κ given β and θ on interval $[\kappa_{1s}, \kappa_{2s}]$ is

$$p_3^s(\kappa|\beta, \theta) = \frac{h_3^{1/2} \mathbf{I}_\kappa([\kappa_{1s}, \kappa_{2s}])}{\int_{\kappa_{1s}}^{\kappa_{2s}} h_3^{1/2} d\kappa} = C_3^s [\psi^{(1)}(\kappa)]^{1/2} \mathbf{I}_\kappa([\kappa_{1s}, \kappa_{2s}]), \quad (8)$$

where C_3^s is a constant depending on κ_{1s} and κ_{2s} , and $\mathbf{I}_x(A)$ is an indicator function that equals 1 if $x \in A$. The obtained conditional prior (8) maximizes the missing information of parameter κ on some compact set. Based on $p_3^s(\kappa|\beta, \theta)$, the conditional prior of (θ, κ) given β on $[\theta_{1s}, \theta_{2s}] \times [\kappa_{1s}, \kappa_{2s}]$ is given by

$$\begin{aligned} p_2^s(\theta, \kappa|\beta) &= \frac{p_3^s(\kappa|\beta, \theta) \exp\left\{\frac{1}{2}E_2^s(\beta, \theta)\right\} \mathbf{I}_\theta([\theta_{1s}, \theta_{2s}])}{\int_{\theta_{1s}}^{\theta_{2s}} \exp\left\{\frac{1}{2}E_2^s(\beta, \theta)\right\} d\theta} \\ &= C_2^s C_3^s \frac{1}{\theta} [\psi^{(1)}(\kappa)]^{1/2} \mathbf{I}_{(\theta, \kappa)}([\theta_{1s}, \theta_{2s}] \times [\kappa_{1s}, \kappa_{2s}]), \end{aligned} \quad (9)$$

where

$$E_2^s(\beta, \theta) = \mathbf{E}_\kappa[\log(h_2)|\beta, \theta] = \int_{\kappa_{1s}}^{\kappa_{2s}} \log(h_2) p_3^s(\kappa|\beta, \theta) d\kappa = \log\left(\frac{\beta^2}{\theta^2}\right),$$

and C_2^s is a constant depending on θ_{1s} and θ_{2s} . The prior in (9) is calculated to maximize the missing information of (θ, κ) on the compact set $[\theta_{1s}, \theta_{2s}] \times [\kappa_{1s}, \kappa_{2s}]$. On the basis of prior $p_2^s(\theta, \kappa|\beta)$, we have the joint prior of $\{\beta, \theta, \kappa\}$ on Ω_s as,

$$p_1^s(\beta, \theta, \kappa) = \frac{p_2^s(\theta, \kappa|\beta) \exp\left\{\frac{1}{2}E_1^s(\beta)\right\} \mathbf{I}_\beta([\beta_{1s}, \beta_{2s}])}{\int_{\beta_{1s}}^{\beta_{2s}} \exp\left\{\frac{1}{2}E_1^s(\beta)\right\} d\beta} = C_1^s C_2^s C_3^s \frac{1}{\beta\theta} [\psi^{(1)}(\kappa)]^{1/2} \mathbf{I}_{(\beta, \theta, \kappa)}(\Omega_s), \quad (10)$$

where

$$E_1^s(\beta) = \mathbf{E}_{\theta, \kappa}[\log(h_1)|\beta] = \int_{\theta_{1s}}^{\theta_{2s}} \int_{\kappa_{1s}}^{\kappa_{2s}} \log(h_1) p_2^s(\theta, \kappa|\beta) d\theta d\kappa = \log\left(\frac{1}{\beta^2}\right),$$

and C_1^s is a constant depending on β_{1s} and β_{2s} . This joint prior (10) maximizes the missing information of the parameters $\{\beta, \theta, \kappa\}$ on the compact set Ω_s . Consequently, the reference prior for $\{\beta, \theta, \kappa\}$ is

$$\pi_1(\boldsymbol{\mu}) = \lim_{s \rightarrow \infty} \frac{p_1^s(\beta, \theta, \kappa)}{p_1^s(\beta_0, \theta_0, \kappa_0)} \propto \frac{1}{\beta\theta} [\psi^{(1)}(\kappa)]^{1/2},$$

where $(\beta_0, \theta_0, \kappa_0) = (1, 1, [\psi^{(1)}(1)]^{-1})$ are specified values of the parameters with $[\psi^{(1)}(x)]^{-1}$ the inverse function of $\psi^{(1)}(x)$.

For the grouping order $\{\theta, \beta, \kappa\}$, an analogous procedure can be implemented with

$$h_1 = \frac{n\beta^2 g(\kappa)}{\theta^2 [nf_1(\kappa)\psi^{(1)}(\kappa) - f_3(\kappa)^2]}, \quad h_2 = \frac{nf_1(\kappa)\psi^{(1)}(\kappa) - f_3(\kappa)^2}{n\beta^2 \psi^{(1)}(\kappa)}, \quad h_3 = n\psi^{(1)}(\kappa).$$

The reference prior of $\{\theta, \beta, \kappa\}$ is obtained as $\pi_1(\boldsymbol{\mu})$. Thus Theorem 1 holds. $\square$

Theorem 2. (a) When the grouping order is $\{\beta, \kappa, \theta\}$, the reference prior of $\boldsymbol{\mu}$ is

$$\pi_2(\boldsymbol{\mu}) \propto \frac{1}{\beta\theta} [\psi^{(1)}(\kappa) - 1/\kappa]^{1/2}, \quad \beta, \theta, \kappa > 0. \quad (11)$$

(b) When the grouping order is $\{\kappa, \beta, \theta\}$, $\{\kappa, \theta, \beta\}$, the reference prior is

$$\pi_3(\boldsymbol{\mu}) \propto \frac{1}{\beta\theta} \left[\frac{g(\kappa)}{n\kappa f_1(\kappa) - f_2(\kappa)^2} \right]^{1/2}, \quad \beta, \theta > 0, \frac{g(\kappa)}{n\kappa f_1(\kappa) - f_2(\kappa)^2} > 0. \quad (12)$$

(c) When the grouping order is $\{\theta, \kappa, \beta\}$, the reference prior is

$$\pi_4(\boldsymbol{\mu}) \propto \frac{1}{\beta\theta} \left[\psi^{(1)}(\kappa) - \frac{f_3(\kappa)^2}{nf_1(\kappa)} \right]^{1/2}, \quad \beta, \theta > 0, \psi^{(1)}(\kappa) - \frac{f_3(\kappa)^2}{nf_1(\kappa)} > 0. \quad (13)$$

The procedures of calculating the reference priors are similar as those in Theorem 1.

Theorems 1 and **2** are about the reference priors of μ for 3 groups. When we partition μ into 2 groups, we focus on patterns $\{(\beta, \theta), \kappa\}$ and $\{\kappa, (\beta, \theta)\}$. This means that parameters β and θ are of equivalent interest, since they are parameters of the intensity function. There are four additional types of 2-group pattern for μ theoretically, such as $\{(\beta, \kappa), \theta\}$, $\{\theta, (\beta, \kappa)\}$, $\{(\theta, \kappa), \beta\}$ and $\{\beta, (\theta, \kappa)\}$. However, we do not take these into consideration for practical reasons. Usually, the intensity function is of interest in practice, which is also determined by both β and θ . Thus, β and θ should be of equal importance.

Theorem 3. (a) When the grouping order is $\{(\beta, \theta), \kappa\}$, the reference prior of μ is

$$\pi_5(\mu) \propto \frac{1}{\theta} [\psi^{(1)}(\kappa)]^{1/2}, \quad \theta, \kappa > 0. \quad (14)$$

(b) When the grouping order is $\{\kappa, (\beta, \theta)\}$, the reference prior of μ is approximately

$$\pi_6(\mu) \propto \frac{1}{\theta} \left[\frac{g(\kappa)}{n\kappa f_1(\kappa) - f_2(\kappa)^2} \right]^{1/2}, \quad \theta > 0, \frac{g(\kappa)}{n\kappa f_1(\kappa) - f_2(\kappa)^2} > 0. \quad (15)$$

Proof. (a) Given grouping order $\{(\beta, \theta), \kappa\}$, we take more interest in (β, θ) than in κ . Following notation in [15], we have

$$h_1 = \begin{pmatrix} \frac{n f_1(\kappa) \psi^{(1)}(\kappa) - f_3(\kappa)^2}{n \beta^2 \psi^{(1)}(\kappa)} & \frac{f_2(\kappa) \psi^{(1)}(\kappa) - f_3(\kappa)}{n \psi^{(1)}(\kappa)} \\ \frac{f_2(\kappa) \psi^{(1)}(\kappa) - f_3(\kappa)}{n \psi^{(1)}(\kappa)} & \frac{n \beta^2 [\psi^{(1)}(\kappa) - 1]}{\theta^2 \psi^{(1)}(\kappa)} \end{pmatrix}, \quad h_2 = n \psi^{(1)}(\kappa).$$

Then the conditional prior of κ given (β, θ) is

$$p_2^s(\kappa | (\beta, \theta)) = \frac{h_2^{1/2}}{\int_{\kappa_{1s}}^{\kappa_{2s}} h_2^{1/2} d\kappa} \mathbf{I}_{\kappa}([\kappa_{1s}, \kappa_{2s}]) \propto [\psi^{(1)}(\kappa)]^{1/2} \mathbf{I}_{\kappa}([\kappa_{1s}, \kappa_{2s}]),$$

and the joint prior of $\{(\beta, \theta), \kappa\}$ is

$$\begin{aligned} p_1^s((\beta, \theta), \kappa) &= \frac{p_2^s(\kappa | (\beta, \theta)) \exp \left\{ \frac{1}{2} \mathbf{E}_{\kappa} [\log |h_1| | (\beta, \theta)] \right\}}{\int_{\beta_{1s}}^{\beta_{2s}} \int_{\theta_{1s}}^{\theta_{2s}} \exp \left\{ \frac{1}{2} \mathbf{E}_{\kappa} [\log |h_1| | (\beta, \theta)] \right\} d\beta d\theta} \mathbf{I}_{(\beta, \theta)}([\beta_{1s}, \beta_{2s}] \times [\theta_{1s}, \theta_{2s}]) \\ &\propto \frac{1}{\theta} [\psi^{(1)}(\kappa)]^{1/2} \mathbf{I}_{\{(\beta, \theta), \kappa\}}(\Omega_s), \end{aligned}$$

where $|h_1| = \frac{g(\kappa)}{\theta^2 \psi^{(1)}(\kappa)}$ is the determinant of matrix h_1 . So that the reference prior of the 2-group pattern $\{(\beta, \theta), \kappa\}$ is

$$\pi_5(\mu) = \lim_{s \rightarrow \infty} \frac{p_1^s((\beta, \theta), \kappa)}{p_1^s((1, 1), [\psi^{(1)}(1)]^{-1})} \propto \frac{1}{\theta} [\psi^{(1)}(\kappa)]^{1/2}.$$

(b) For grouping order $\{\kappa, (\beta, \theta)\}$,

$$\begin{aligned} h_1 &= \frac{ng(\kappa)}{n\kappa f_1(\kappa) - f_2(\kappa)^2}, \quad h_2 = \begin{pmatrix} \frac{f_1(\kappa)}{\beta^2} & \frac{f_2(\kappa)}{\theta} \\ \frac{f_2(\kappa)}{\theta} & \frac{n\kappa \beta^2}{\theta^2} \end{pmatrix}, \quad |h_2| = \frac{n\kappa f_1(\kappa) - f_2(\kappa)^2}{\theta^2}. \\ p_2^s((\beta, \theta) | \kappa) &= \frac{|h_2|^{1/2}}{\int_{\beta_{1s}}^{\beta_{2s}} \int_{\theta_{1s}}^{\theta_{2s}} |h_2|^{1/2} d\beta d\theta} \mathbf{I}_{(\beta, \theta)}([\beta_{1s}, \beta_{2s}] \times [\theta_{1s}, \theta_{2s}]) \propto \frac{1}{\theta} \mathbf{I}_{(\beta, \theta)}([\beta_{1s}, \beta_{2s}] \times [\theta_{1s}, \theta_{2s}]), \\ p_1^s(\kappa, (\beta, \theta)) &= \frac{p_2^s((\beta, \theta) | \kappa) \exp \left\{ \frac{1}{2} \mathbf{E}_{\beta, \theta} [\log(h_1) | \kappa] \right\}}{\int_{\kappa_{1s}}^{\kappa_{2s}} \exp \left\{ \frac{1}{2} \mathbf{E}_{\beta, \theta} [\log(h_1) | \kappa] \right\} d\kappa} \mathbf{I}_{\kappa}([\kappa_{1s}, \kappa_{2s}]) \\ &\propto \frac{1}{\theta} \left[\frac{g(\kappa)}{n\kappa f_1(\kappa) - f_2(\kappa)^2} \right]^{1/2}. \end{aligned}$$

The reference prior of the 2-group pattern $\{\kappa, (\beta, \theta)\}$ is thus

$$\lim_{s \rightarrow \infty} \frac{p_1^s(\kappa, (\beta, \theta))}{p_1^s(\kappa_0, (1, 1))} \propto \frac{1}{\theta} \left[\frac{g(\kappa)}{n\kappa f_1(\kappa) - f_2(\kappa)^2} \right]^{1/2},$$

which is exactly $\pi_6(\mu)$. $\square$

According to **Theorems 1–3**, six reference priors are obtained for different grouping orders of the parameters. When we include the Jeffreys prior, we have the seven listed in **Table 1**. For reference priors $\pi_3(\mu)$, $\pi_4(\mu)$ and $\pi_6(\mu)$, they set undesirable constraints on parameter κ . However, notice that

$$\frac{g(\kappa)}{n\kappa f_1(\kappa) - f_2(\kappa)^2} = \psi^{(1)}(\kappa) - \frac{n f_1(\kappa) - 2 f_2(\kappa) f_3(\kappa) + \kappa f_3(\kappa)^2}{n\kappa f_1(\kappa) - f_2(\kappa)^2},$$

Table 1
Possible reference priors.

Grouping order	Reference prior
$\{\beta, \theta, \kappa\}, \{\theta, \beta, \kappa\}$	$\pi_1(\boldsymbol{\mu}) \propto \frac{1}{\beta\theta} [\psi^{(1)}(\kappa)]^{1/2}$
$\{\beta, \kappa, \theta\}$	$\pi_2(\boldsymbol{\mu}) \propto \frac{1}{\beta\theta} [\psi^{(1)}(\kappa) - 1/\kappa]^{1/2}$
$\{\kappa, \beta, \theta\}, \{\kappa, \theta, \beta\}$	$\pi_3(\boldsymbol{\mu}) \propto \frac{1}{\beta\theta} \left[\frac{g(\kappa)}{n\kappa f_1(\kappa) - f_2(\kappa)^2} \right]^{1/2}$
$\{\theta, \kappa, \beta\}$	$\pi_4(\boldsymbol{\mu}) \propto \frac{1}{\beta\theta} \left[\psi^{(1)}(\kappa) - \frac{f_3(\kappa)^2}{n f_1(\kappa)} \right]^{1/2}$
$\{(\beta, \theta), \kappa\}$	$\pi_5(\boldsymbol{\mu}) \propto \frac{1}{\theta} [\psi^{(1)}(\kappa)]^{1/2}$
$\{\kappa, (\beta, \theta)\}$	$\pi_6(\boldsymbol{\mu}) \propto \frac{1}{\theta} \left[\frac{g(\kappa)}{n\kappa f_1(\kappa) - f_2(\kappa)^2} \right]^{1/2}$
$\{(\beta, \theta, \kappa)\}$	$\pi_7(\boldsymbol{\mu}) \propto \frac{1}{\theta} [g(\kappa)]^{1/2}$

and for $\kappa > 0$,

$$\lim_{n \rightarrow \infty} \frac{n f_1(\kappa) - 2 f_2(\kappa) f_3(\kappa) + \kappa f_3(\kappa)^2}{n \kappa f_1(\kappa) - f_2(\kappa)^2} = 1/\kappa. \quad (16)$$

Thus,

$$\lim_{n \rightarrow \infty} \pi_3(\boldsymbol{\mu}) = \pi_2(\boldsymbol{\mu}) \quad \text{and} \quad \lim_{n \rightarrow \infty} \pi_6(\boldsymbol{\mu}) \propto \frac{1}{\theta} [\psi^{(1)}(\kappa) - 1/\kappa]^{1/2}.$$

For the convenience of application, substitute $\pi_2(\boldsymbol{\mu})$ and $\frac{1}{\theta} [\psi^{(1)}(\kappa) - 1/\kappa]^{1/2}$ for $\pi_3(\boldsymbol{\mu})$ and $\pi_6(\boldsymbol{\mu})$ respectively. When the parameters in the intensity function are our interested parameters, then the prior of $\pi_1(\boldsymbol{\mu})$ is more suitable for our analysis. If the analyst is mainly interested in understanding the measure of improvement or deterioration over the system's life and the repair effect, the more interesting parameters being β and κ , not θ , then the prior of $\pi_2(\boldsymbol{\mu})$ is more suitable for our analysis. When the parameter of the measure of improvement or deterioration over the system's life and the scale parameter are our interested parameters simultaneously, then the prior of $\pi_5(\boldsymbol{\mu})$ is more suitable for our analysis. If the parameter of repair effect is of greatest interest, then the priors $\pi_3(\boldsymbol{\mu})$ and $\pi_6(\boldsymbol{\mu})$ are more suitable for our analysis. In real application, the scale parameter θ is not our most interested parameter in general, so the prior of $\pi_4(\boldsymbol{\mu})$ is not used often. Consequently, in this article, we will utilize four reference priors which include $\pi_1(\boldsymbol{\mu})$, $\pi_2(\boldsymbol{\mu})$, $\pi_5(\boldsymbol{\mu})$, $\lim_{n \rightarrow \infty} \pi_3(\boldsymbol{\mu})$ and $\lim_{n \rightarrow \infty} \pi_6(\boldsymbol{\mu})$ to proceed the Bayesian posterior analysis. The four are $\frac{1}{\beta\theta} [\psi^{(1)}(\kappa)]^{1/2}$, $\frac{1}{\beta\theta} [\psi^{(1)}(\kappa) - 1/\kappa]^{1/2}$, $\frac{1}{\theta} [\psi^{(1)}(\kappa)]^{1/2}$ and $\frac{1}{\theta} [\psi^{(1)}(\kappa) - 1/\kappa]^{1/2}$.

4. Posterior analysis

For the sake of convenience, we write the four reference priors in a general way:

$$\pi_{ab}(\boldsymbol{\mu}) \propto \frac{1}{\beta^a \theta^b} [\psi^{(1)}(\kappa) - b/\kappa]^{1/2}, \quad a, b = 0, 1. \quad (17)$$

$\pi_{ab}(\boldsymbol{\mu})$ covers three possibilities in Table 1, namely, $\pi_1(\boldsymbol{\mu})$, $\pi_2(\boldsymbol{\mu})$, and $\pi_5(\boldsymbol{\mu})$ and $\pi_{01}(\boldsymbol{\mu})$ equals to $\lim_{n \rightarrow \infty} \pi_6(\boldsymbol{\mu})$. According to Bayes formula, the joint posterior distribution of $\boldsymbol{\mu}$ based on $\pi_{ab}(\boldsymbol{\mu})$ is

$$\pi_{ab}(\boldsymbol{\mu} | \mathbf{t}) = \frac{L(\mathbf{t} | \boldsymbol{\mu}) \pi_{ab}(\boldsymbol{\mu})}{\int_0^\infty \int_0^\infty \int_0^\infty L(\mathbf{t} | \boldsymbol{\mu}) \pi_{ab}(\boldsymbol{\mu}) d\beta d\theta d\kappa},$$

where the denominator $D = \int_0^\infty \int_0^\infty \int_0^\infty L(\mathbf{t} | \boldsymbol{\mu}) \pi_{ab}(\boldsymbol{\mu}) d\beta d\theta d\kappa$ is nominal constant of the posterior density. After integrating over θ of the denominator D , we get

$$D = \int_0^\infty \int_0^\infty \frac{\Gamma(n\kappa)}{\Gamma(\kappa)^n} \beta^{n-a-1} t_n^{n\kappa\beta} [\psi^{(1)}(\kappa) - b/\kappa]^{1/2} \prod_{i=1}^n (t_i^\beta - t_{i-1}^\beta)^{\kappa-1} d\beta d\kappa.$$

To conduct valid Bayesian analysis for the model, propriety of the posterior distribution is indispensable; see following theorem.

Theorem 4. For $\beta > 0$, $\theta > 0$ and $\kappa > 0$, the joint posterior distribution of $\boldsymbol{\mu}$ is proper based on prior $\pi_{ab}(\boldsymbol{\mu})$ if sample size $n > a$.

Proof. The proof is very complicated and we show it in the appendix. $\square$

Theorem 4 guarantees the Bayesian inference based on the reference priors are valid. In order to obtain the estimate of $\boldsymbol{\mu} = (\beta, \theta, \kappa)$, Gibbs sampling procedure can be implemented with the full conditional posteriors of the components of $\boldsymbol{\mu}$ given as follows.

1. With the joint posterior distribution given in the appendix, we derive that

$$\theta^{-\beta} | \mathbf{t}, \beta, \kappa \sim \text{Gamma}\left(n\kappa + \frac{1}{\beta} + 1, t_n^\beta\right). \quad (18)$$

This gives rise to the full conditional distribution of θ .

2. The full conditional posterior of β is obtained as,

$$\pi_{ab}(\beta|\mathbf{t}, \theta, \kappa) \propto \frac{\beta^{n-a}}{\theta^{n\beta\kappa}} \left(\prod_{i=1}^n t_i \right)^{\beta} \left[\prod_{i=1}^n (t_i^{\beta} - t_{i-1}^{\beta}) \right]^{\kappa-1} \exp \left\{ -\left(\frac{t_n}{\theta} \right)^{\beta} \right\}. \quad (19)$$

The second derivative of $\log \pi(\beta|\mathbf{t}, \theta, \kappa)$ with respect to β is

$$-\frac{n-a}{\beta^2} - \left(\frac{t_n}{\theta} \right)^{\beta} \left(\log \frac{t_n}{\theta} \right)^2 - (\kappa-1) \sum_{i=2}^n \frac{t_{i-1}^{\beta} t_i^{\beta} (\log t_i - \log t_{i-1})^2}{(t_i^{\beta} - t_{i-1}^{\beta})^2} < 0.$$

Then $\pi_{ab}(\beta|\mathbf{t}, \theta, \kappa)$ is log-concave and adaptive rejection sampling [18] can be used to generate β .

3. For $\kappa > 0$, the full conditional posterior is given by

$$\pi_{ab}(\kappa|\mathbf{t}, \beta, \theta) \propto \frac{1}{\Gamma(\kappa)^n \theta^{n\beta\kappa}} \left[\prod_{i=1}^n (t_i^{\beta} - t_{i-1}^{\beta}) \right]^{\kappa} [\psi^{(1)}(\kappa) - b/\kappa]^{1/2}. \quad (20)$$

The second derivative of $\log \pi(\kappa|\mathbf{t}, \beta, \theta)$ with respect to κ is

$$-n\psi^{(1)}(\kappa) + \frac{[\kappa^3 \psi^{(3)}(\kappa) - 2b][\kappa \psi^{(1)}(\kappa) - b] - [\kappa^2 \psi^{(2)}(\kappa) + b]^2}{2[\kappa \psi^{(1)}(\kappa) - b]^2} < 0 \text{ for any } n \geq 1.$$

Accordingly, $\pi_{ab}(\kappa|\mathbf{t}, \beta, \theta)$ is also log-concave and the adaptive rejection sampling can also be used to generate κ .

Using Gibbs sampling, we are able to get posterior samples of the parameters. Bayesian inference therefore is carried out completely depending on them.

5. Bayesian prediction

One of the goals for analyzing repairable system is to get insights into its failure pattern, so that future failure times, as well as cost and loss, can be predicted. Let t_{n+m} be the m th future failure time of the repairable system. Then the predicted likelihood function of it is given as,

$$f(t_{n+m}|\mathbf{t}_n; \boldsymbol{\mu}) = \frac{1}{\Gamma(m\kappa)} \frac{\beta}{\theta} \left(\frac{t_{n+m}}{\theta} \right)^{\beta-1} \left[\left(\frac{t_{n+m}}{\theta} \right)^{\beta} - \left(\frac{t_n}{\theta} \right)^{\beta} \right]^{m\kappa-1} \exp \left\{ -\left(\frac{t_{n+m}}{\theta} \right)^{\beta} + \left(\frac{t_n}{\theta} \right)^{\beta} \right\}, \quad t_{n+m} > t_n, \quad (21)$$

The posterior predictive distribution of t_{n+m} , which is marginalized over the posterior $\pi_{ab}(\boldsymbol{\mu}|\mathbf{t})$, is

$$\begin{aligned} f_{ab}(t_{n+m}|\mathbf{t}_n) &= \int_0^\infty \int_0^\infty \int_0^\infty f(t_{n+m}|\mathbf{t}_n; \boldsymbol{\mu}) \pi_{ab}(\boldsymbol{\mu}|\mathbf{t}) d\beta d\theta d\kappa \\ &= \frac{1}{D} \int_0^\infty \int_0^\infty \frac{\Gamma(\kappa(m+n))}{\Gamma(m\kappa)[\Gamma(\kappa)]^n} \frac{\beta^{n-a}}{t_{n+m}^{\beta[\kappa(n+m)-1]+1}} \prod_{i=1}^n t_i^{\beta-1} \prod_{i=1}^n (t_i^{\beta} - t_{i-1}^{\beta})^{\kappa-1} \\ &\quad \times (t_{n+m}^{\beta} - t_n^{\beta})^{m\kappa-1} [\psi^{(1)}(\kappa) - b/\kappa]^{1/2} d\beta d\kappa, \quad t_{n+m} > t_n, \end{aligned} \quad (22)$$

and $\int_{t_n}^\infty f_{ab}(t_{n+m}|\mathbf{t}_n) dt_{n+m} = 1$. To draw samples of t_{n+m} from (22), which forms the basis for inference for future failure times, we apply the method of composition [19]. The procedures are listed as follows:

1. Generate $\boldsymbol{\mu}^*$ from $\pi_{ab}(\boldsymbol{\mu}|\mathbf{t})$.
2. Based on $\boldsymbol{\mu}^*$, generate t_{n+m}^* from $f_{ab}(t_{n+m}|\mathbf{t}_n; \boldsymbol{\mu}^*)$.

Steps 1 and 2 are repeated several times to yield the desired sample. Posterior distributions of any functional of the future observation are computed by using these samples of t_{n+m} .

6. Simulation and application

6.1. Simulation studies

In this section, the results of simulation are presented to evaluate the performance of the proposed four reference priors ($\pi_{ab}(\boldsymbol{\mu})$, $a, b = 0, 1$). We consider failure-truncated cases of different sample sizes: $n = 15$ and $n = 30$. The true values of parameters are assumed to be 1.5 (> 1), $\theta = 2$ and $\kappa = 3$. Based on the pseudo-random MPLP samples and the four reference priors, Gibbs sampling can be implemented to generate corresponding posterior samples of the parameters with their full conditional posterior distributions respectively given by (18)–(20) for different values of a and b .

In our simulation studies, a total number of 1000 failure-truncated pseudo-random MPLP samples are repeatedly simulated under each data configuration. For each simulated MPLP sample, the iteration number of Gibbs sampling is 5000

Table 2Parameter estimates with true values $(\beta, \theta, \kappa) = (1.5, 2, 3)$.

Bayes or MLE	Para.	$n = 15$			$n = 30$		
		Bias	RMSE	CP (95%)	Bias	RMSE	CP (95%)
$\pi_1(\mu)$	β	0.0387	0.2533	0.933	0.0173	0.1580	0.941
	θ	0.2611	1.0945	0.932	0.1839	0.7891	0.952
	κ	0.7023	1.7698	0.943	0.2430	0.8754	0.949
$\pi_2(\mu)$	β	0.0423	0.2557	0.949	0.0173	0.1586	0.948
	θ	0.3688	1.1596	0.942	0.2268	0.8138	0.950
	κ	0.4771	1.5948	0.943	0.1501	0.8278	0.951
$\pi_3(\mu)$	β	0.0785	0.2726	0.931	0.0354	0.1632	0.953
	θ	0.4129	1.2182	0.935	0.2639	0.8381	0.943
	κ	0.6935	1.7942	0.943	0.2361	0.8695	0.957
$\pi_4(\mu)$	β	0.0843	0.2754	0.938	0.0372	0.1641	0.939
	θ	0.5300	1.2940	0.937	0.3162	0.8733	0.943
	κ	0.4701	1.6033	0.943	0.1427	0.8234	0.948
MLE	β	0.0604	0.2626	0.910	0.0310	0.1605	0.945
	θ	0.0452	1.0375	0.855	0.0770	0.7546	0.931
	κ	0.9543	1.9892	0.964	0.3469	0.9313	0.960

Table 3Coverage rates of 2-sided Bayesian 90% and 95% prediction intervals on t_{n+m} with $n = 30$ for different data configurations.

(β, θ, κ)	Priors	90% prediction interval			95% prediction interval		
		$m = 1$	$m = 3$	$m = 5$	$m = 1$	$m = 3$	$m = 5$
(1.5,2,1.5)	$\pi_1(\mu)$	0.894	0.883	0.907	0.955	0.940	0.948
	$\pi_2(\mu)$	0.902	0.888	0.907	0.958	0.943	0.952
	$\pi_3(\mu)$	0.893	0.884	0.953	0.946	0.937	0.970
	$\pi_4(\mu)$	0.899	0.891	0.908	0.953	0.940	0.951
	$\pi_1(\mu)$	0.893	0.891	0.916	0.941	0.949	0.946
(1.5,2,3)	$\pi_2(\mu)$	0.886	0.894	0.913	0.947	0.943	0.945
	$\pi_3(\mu)$	0.888	0.891	0.968	0.943	0.945	0.976
	$\pi_4(\mu)$	0.893	0.893	0.909	0.944	0.952	0.948
	$\pi_1(\mu)$	0.908	0.912	0.902	0.960	0.955	0.951
	$\pi_2(\mu)$	0.915	0.913	0.901	0.958	0.958	0.952
(0.75,2,1.5)	$\pi_3(\mu)$	0.905	0.910	0.946	0.955	0.954	0.970
	$\pi_4(\mu)$	0.910	0.912	0.901	0.959	0.958	0.951
	$\pi_1(\mu)$	0.905	0.895	0.872	0.951	0.947	0.936
	$\pi_2(\mu)$	0.907	0.898	0.882	0.954	0.955	0.936
	$\pi_3(\mu)$	0.903	0.885	0.933	0.950	0.946	0.960
(0.75,2,3)	$\pi_4(\mu)$	0.910	0.908	0.885	0.951	0.951	0.938

with the first 3000 samples as burn-in samples. The software R took about 4–6 s on a HP PC in each sampling. Under the squared error loss, Bayesian estimates (BE) of the parameters are their expectations of the corresponding marginal posterior distributions. Thus BEs can be obtained via the means of posterior samples. For the purpose of comparison, classical maximum likelihood estimates (MLE) are also taken into account. Absolute biases of the estimates, rooted mean squared errors (RMSE) and 95% coverage probabilities (CP) under four reference priors for Bayesian analysis and maximum likelihood approach are calculated. For the maximum likelihood method, the asymptotic distribution of the estimator $(\hat{\theta}, \hat{\beta}, \hat{\kappa})$ is multivariate normal with mean (θ, β, κ) and covariance matrix $[\mathbf{I}(\theta, \beta, \kappa)]^{-1}$, where the $\mathbf{I}$ matrix is given in (5). This asymptotic normal distribution is used to construct the approximate confidence intervals. The coverage probability for Bayesian analysis is defined as the fraction of times an equal tail 2-sided $1 - \alpha$ ($\alpha = 0.05$) Bayesian credible interval covers the true value of the parameter in repeated sampling. The simulation results are shown in Tables 2.

In Table 2, for the settings where true value $\beta = 1.5$ (>1), both Bayesian and maximum likelihood approaches give accurate point estimates and appropriate RMSEs of the parameters. However, the former one provides so much better performance than the latter one in terms of meeting the target coverage probabilities, even under the situation where sample size is small! For the Bayesian method, although the four reference priors behave equally well, we can still see that the 95% CPs from $\pi_2(\mu)$ are more satisfactory for $n = 15$ and $n = 30$.

In order to compare our method with Black and Rigdon [10] and Calabria and Pulcini [11], the coverage rates of the Bayesian credibility intervals have been evaluated for $1 - \alpha = 95\%$ and for $\kappa = 1, 2, 3$. In Table 4, the Bayesian results with the prior $\pi_2(\mu)$ proposed in our paper are given and are compared with the method proposed by Black and Rigdon [10] and Calabria and Pulcini [11]. It appears that the Bayesian procedure proposed by Calabria and Pulcini [11] and our paper outperforms the ML one proposed by Black and Rigdon [10] since the coverage rate of the Bayesian 95% credibility intervals is closer to 95% than the ML ones. Besides, the coverage rate with the prior $\pi_2(\mu)$ is slightly better than the prior of

Table 4

A comparative of the results of coverage rates of Bayes 95% credibility intervals and of ML 95% confidence intervals on β and κ between of Black and Rigdon [10], Calabria and Pulcini [11] and our method.

Sample size n	Black and Rigdon				Calabria and Pulcini				Reference prior	
	True value of		Results		$g(\kappa) \propto [\psi'(\kappa)]^{1/2}$		$g(\kappa) \propto \kappa^{-1}$		$\pi_2(\mu)$	
	β	κ	β	κ	β	κ	β	κ	β	κ
10	1.00	1	92.4%	97.6%	94.2%	93.1%	95.1%	94.7%	94.3%	94.8%
	1.00	2	88.4%	98.0%	94.1%	93.6%	94.9%	94.4%	95.3%	94.6%
	1.00	3	90.1%	98.5%	93.3%	93.8%	94.5%	95.3%	94.8%	94.7%
20	1.00	1	93.8%	96.6%	95.5%	94.9%	95.8%	95.6%	94.7%	94.6%
	1.00	2	93.3%	96.0%	93.4%	93.4%	93.8%	94.2%	95.4%	95.6%
	1.00	3	91.5%	96.9%	94.7%	94.5%	95.0%	95.1%	94.5%	95.4%

$\beta^{-1}\theta^{-1}[\psi'(\kappa)]^{1/2}$ proposed in [11]. Because the coverage rates based on reference prior are slightly closer to the nominal level.

From a perspective of prediction, the maximum likelihood method cannot provide a solution to the problem, whereas Bayesian method can predict the future failure time t_{n+m} on the basis of samples arising from (22). The procedure for drawing predictive samples is listed in Section 5. In the Monte Carlo simulation study, based on 1000 failure-truncated MPLP samples of size $n = 15$ and $n = 30$ respectively, the coverage probability of the Bayes predictive interval is computed which is defined by the fraction of times the 2-sided Bayes predictive interval on t_{n+m} covers the (simulated) predicted observation in repeated sampling. The simulation results are given in Table 3 for $m = 1, 3, 5$ and $1 - \alpha = 90\%$ and 95% . The results show that Bayesian predictive intervals based on reference priors $\pi_5(\mu)$ do not possess as good statistical properties as those based on other three reference priors in a frequentist context. The coverage probabilities arising from $\pi_5(\mu)$ for $m = 5$ always overestimate the nominal levels. Hence, $\pi_1(\mu)$, $\pi_2(\mu)$ and $\lim_{n \rightarrow \infty} \pi_6(\mu)$ are better choices in predicting the future failure time.

From the estimation and prediction results shown above, we conclude that objective Bayes outperforms the maximum likelihood approach in assessing the parameters and obtaining the target CPs. Moreover, among the four reference priors in Bayesian analysis, $\pi_2(\mu) = \frac{1}{\beta\theta}[\psi^{(1)}(\kappa) - 1/\kappa]^{1/2}$ is superior to others which therefore will be our recommendation to researchers. To illustrate convergence of Bayesian posterior samples of the four reference priors, we give the trace plots of the parameters with burn-in samples in Fig. 1. Fig. 1 demonstrates the trace plots after discarding the burn-in samples are not affected significantly by starting value $\mu^{(0)} = (1, 1, 1)$. And the trace plots of the parameters show the posterior samples fluctuate near the true value when the first 3000 burn-in samples are removed which shows the samples convergence to a stationary distribution.

6.2. Real data analysis

6.2.1. US business cycle data

We study the US business cycle data from National Bureau of Economic Research (NBER). The data set may not come from a “repairable system”, but the data pattern meets the requirement of modulated power law process that will be confirmed in the following content. The data are measured in months, from December 1854 to June 2009, including 33 observations of US business contraction and expansion. The contractions (recessions) start at the peak of a business cycle and end at the trough. The expansions start at the trough and end at the peak. The dates of peak and trough are announced by the Business Cycle Dating Committee of NBER. The determination that the last expansion began in June 2009 is the most recent decision of the committee, dated on September 20, 2010. The dates and duration of US business cycle contractions and expansions is available at <http://www.nber.org/cycles.html/>.

In the data set, both contraction and expansion can be treated as recurrent events and they occurred alternatively. In this example, the unit of time t_i and θ is in months. For contraction, assume that the peak date June 1857 is the start point $t_0 = 0$. The first contraction therefore occurred at $t_1 = 18$; the second one occurred at $t_2 = 18 + 8 = 26$; the third one occurred at $t_3 = 18 + 8 + 32 = 58$ and so as the next. For expansion, December 1854 is assumed to be the start point. The first expansion occurred at $t_1 = 30$ accordingly; the second expansion occurred at $t_2 = 30 + 22 = 52$ and so on. Fig. 2 illustrates the occurrence times of 33 US business contractions and expansions.

Fig. 2 indicate that US business cycles are not uniform in length. Contractions during the period covered have been as short as 6 months and as long as 65 months, while expansions varied from 10 to 120 months. Slopes of contraction and expansion time plots in Fig. 2, depicting the occurrence rates of them, are not constant. In particular, the occurrence rate of contraction has become larger since World War II (WWII), while that of expansion has become smaller. In other words, the duration of contraction has become shorter since WWII, while that of expansion has become longer. All these suggest that US business cycles have changed over time. Hence, the modulated power law process can be applied to model the US business data. Three parameters, β , θ and κ , of the MPLP model are estimated considering Bayesian reference prior $\pi_2(\mu)$. To show the efficiency of Bayesian prediction, we utilize only the first 32 observations (from December 1854 to November

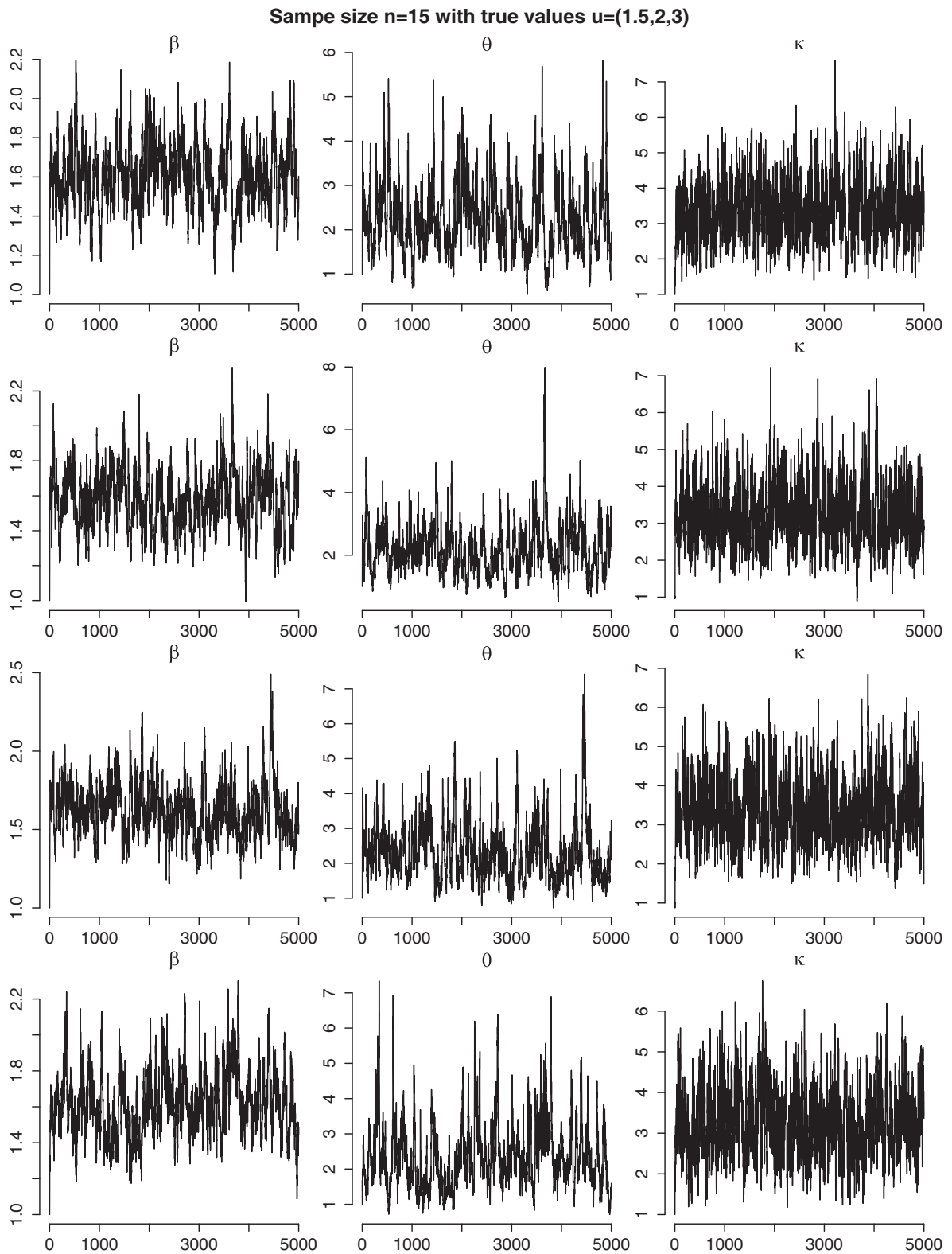


Fig. 1. Trace of posterior samples of (β, θ, κ) with $n = 15$ and true values $\mu = (1.5, 2, 3)$.

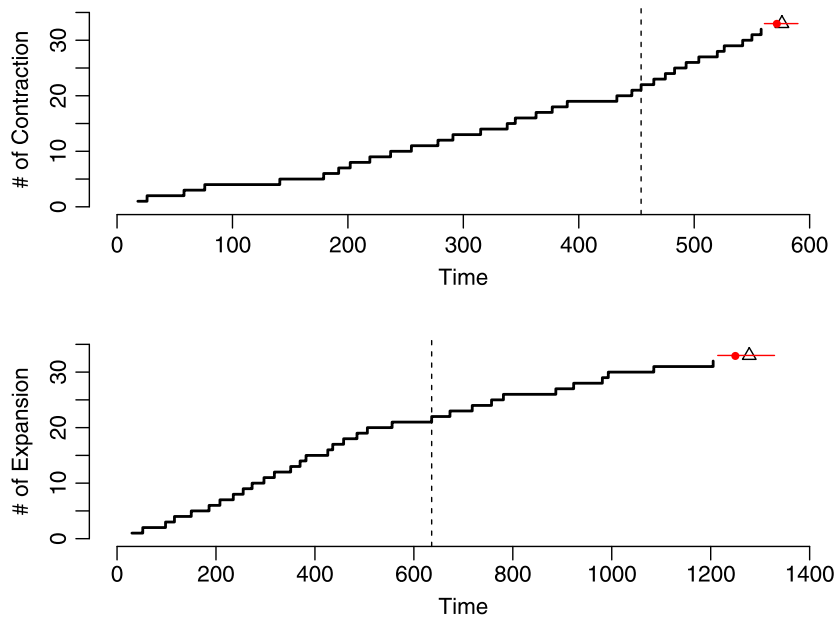


Fig. 2. Occurrence times of 33 US business contractions and expansions. The vertical dashed line is used to distinguish between contractions (expansion) pre-WWII and post-WWII. The triangle and dot respectively represent the true and predictive occurrence time of the 33rd contraction (expansion). The horizontal line shows the 95% predictive interval for the 33rd contraction (expansion) time.

Table 5

Point estimation and test of the MPLP parameters.

	Bayes	MLE
<i>Results of contraction data</i>		
β	1.2288 [0.1232]	1.2658 [0.1211]
θ	13.3381 [6.1340]	13.4358 [5.6353]
κ	3.3903 [0.8167]	3.4913 [0.8316]
Bayes factor B_{01}/p -Value: $H_0 : \kappa = 1$	<0.0001	<0.0001
Bayes factor B_{01}/p -Value: $H_0 : \beta = 1$	0.0182(< 1:30)	0.0053(< 0.05)
Bayes factor B_{01}/p -Value: $H_0 : \kappa = \beta = 1$	<0.0001	<0.0001
Prediction of the 33rd duration	13.8865 [7.6665]	–
95% predictive interval	(2.8183, 31.8856)	–
<i>Results of expansion data</i>		
β	0.7813 [0.0738]	0.8056 [0.0804]
θ	4.0639 [2.8273]	3.8311 [2.5778]
κ	3.0886 [0.7045]	3.2152 [0.7671]
Bayes factor B_{01}/p -Value: $H_0 : \kappa = 1$	0.0035(< 1:100)	<0.0001
Bayes factor B_{01}/p -Value: $H_0 : \beta = 1$	0.8892(< 1)	0.0145(< 0.05)
Bayes factor B_{01}/p -Value: $H_0 : \kappa = \beta = 1$	0.0041(< 1:100)	<0.0001
Prediction of the 33rd duration	49.3176 [29.7390]	–
95% predictive interval	(10.2044, 123.0641)	–

The table shows Bayesian and maximum likelihood estimates of β , θ and κ , Bayes factor and p -Value for hypothesis test and Bayesian prediction for the 33rd duration of contraction and expansion. The standard deviations of estimates are reported in square brackets which is reached through asymptotic normal distribution. NBER business cycle data from December 1854 to November 2001 are used.

2001) of US business contraction and expansion to make inference, and the 33rd one is used in prediction and comparative study.

Table 5 reports the Bayesian estimates and maximum likelihood estimates (MLE) for the contraction data and expansion data, respectively. Estimation results from the two approaches are similar. For the Bayesian estimates, the iteration number of Gibbs sampling is 5000 with the first 3000 samples as burn-in samples. The run time for the Bayesian estimates is 7.786 s and the run time for the maximum likelihood estimates (MLE) is 0.419 s. For contraction data, the estimated shape parameter β is near 1.2(>1), implying that the contractions are getting shorter and shorter. For expansion data, the estimated β is near 0.8(<1), indicating that the duration of expansion is becoming longer and longer. Those are consistent with the finding in Fig. 2. Here the standard error of MLE is the squared root of the diagonal elements of inverse fisher information matrix. The standard error of Bayesian estimates is the standard deviation of posterior sample.

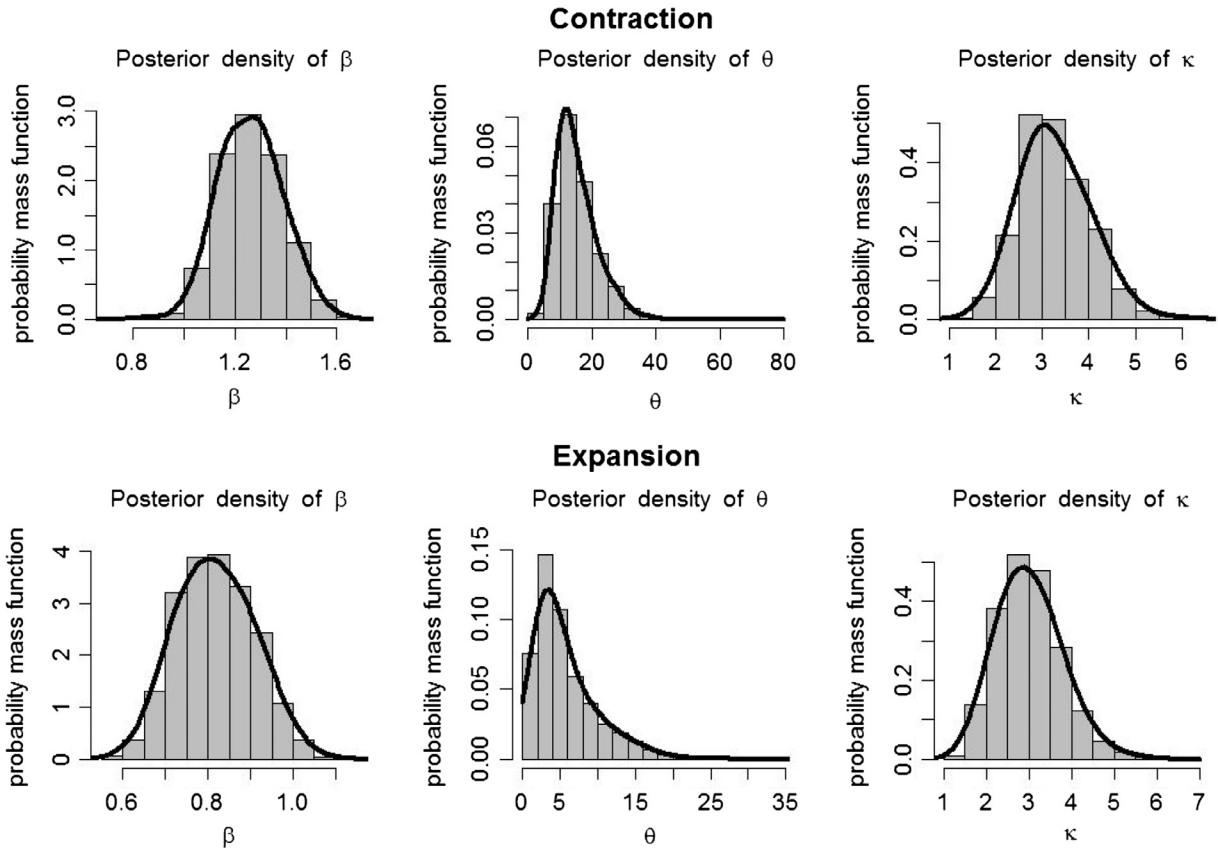


Fig. 3. Bayesian posterior densities of β , θ and κ considering reference prior $\pi_2(\mu)$ for US business contraction and expansion.

In order to test for a reduction of the MPLP to simpler models, Bayesian hypothesis testing is applied. We consider three hypotheses: $H_0 : \kappa = 1$ (NHPP), $H_0 : \beta = 1$ (Gamma RP) and $H_0 : \kappa = \beta = 1$ (HPP) versus $H_1 : \beta, \theta > 0, \kappa > 1$ (MPLP) respectively. The priors of parameters in testing are on the basis of $\pi_2(\mu)$ and the prior odds of H_0 to H_1 is assumed to be 1. Bayes factors of H_0 to H_1 , say B_{01} , are calculated and listed in Table 5. The p -Values based on likelihood ratio test are also computed for reference. For Bayes factors, Jeffereys [20] suggested that there was decisive evidence to reject H_0 if $B_{01} < 1 : 100$, very strong evidence to reject it if $1 : 100 < B_{01} < 1 : 30$, strong evidence to reject it if $1 : 30 < B_{01} < 1 : 10$, and barely worth mentioning evidence to reject it if $1 : 3 < B_{01} < 1$. From the testing results, we see that B_{01} of $H_0 : \beta = 1$ in expansion data is 0.8892 which may be a little bit helpful for decision making. There is no evidence against the hypothesis that $\beta = 1$.

Although Bayesian and maximum likelihood methods perform equally well, the former one has its superiority in predicting. Point and 95% interval prediction of duration of the 33rd US business contraction and expansion are also shown in Table 5. Here the point prediction is the posterior mean. For contraction, point prediction of the 33rd duration, 13.89, is close to the true value 18. For expansion, on the other hand, the point prediction 49.32 is a little far away from the true value 73. That is, the true expansion has occurred two years later than the time we predict. This might be due to the larger volatility and a more dispersed distribution of the duration of expansion in the given period. Nevertheless, true duration values of contraction and expansion are in the corresponding 95% predictive intervals, respectively. Fig. 2 illustrates our predictive points and 95% predictive intervals for the 33rd duration. Histograms and approximate probability density functions for the predictive distribution of the 33rd duration for US business contraction and expansion are shown in Figs. 3 and 4.

6.2.2. Aircraft generator failure data

Let us consider the failure times of an aircraft air generator taken from Black and Rigdon [10], which is a repairable system data. In this example, the unit of time t_i and θ is in hours. Table 6 shows the failure times and this data set was analyzed by Black and Rigdon [10] and Calabria and Pulcini [11]. Black and Rigdon [10] obtained ML point estimators and 95% confidence intervals of β , θ and κ . Calabria and Pulcini [11] obtained point estimators by the posterior mean and posterior median and 95% credibility interval. Na et al. [21] proposed Bayesian estimation of the next failure time after the system has experienced some failures for the MPLP model. The asymptotic normal approximation for the ML estimators of β and κ is used to produce confidence intervals and the logarithmic transformation of the ML estimator introduced in [10] is used to produce confidence interval on θ . The comparison results are listed in Table 7. In Table 7, the results of

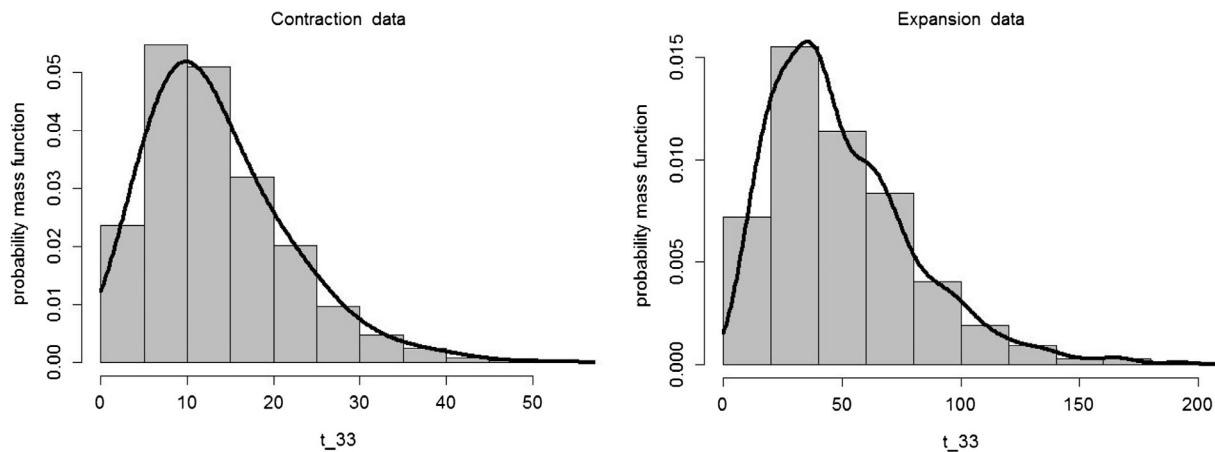


Fig. 4. Predictive pdf of the 33rd duration for US business contraction and expansion.

Table 6
Aircraft generator failure time data.

Failure number	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Failure time	10	55	166	205	341	488	567	731	1308	2050	2453	3115	4017	4596

The table shows the failure times of aircraft generator.

Table 7
The comparison results among different methods.

	β			θ			κ		
	ML	Bayes 1997	Bayes ($\pi_2(\mu)$)	ML	Bayes 1997	Bayes ($\pi_2(\mu)$)	ML	Bayes 1997	Bayes ($\pi_2(\mu)$)
Point estimators	0.42	0.419(*) 0.415($\diamond$)	0.423(*) 0.420($\diamond$)	0.218	1.30(*) 0.272($\diamond$)	0.82(*) 0.266($\diamond$)	4.8	4.2(*) 3.98($\diamond$)	4.56(*) 4.28($\diamond$)
Lower limit	0.32	0.307	0.317	0.009	0.008	0.006	1.34	1.66	1.73
Upper limit	0.53	0.553	0.564	5.29	8.69	8.53	8.26	7.98	8.19

(*) Denotes the posterior mean, ($\diamond$) denotes the posterior median.

Table 8
Point estimation and test of the MPLP parameters.

	Bayes	MLE
β	0.4230 [0.0439]	0.4223 [0.0540]
θ	0.8291 [1.0650]	0.2164 [0.3587]
κ	4.5610 [1.5258]	4.8033 [1.7823]
Bayes factor B_{01}/p -Value: $H_0 : \kappa = 1$	0.0321(< 1:30)	0.0083(< 0.05)
Bayes factor B_{01}/p -Value: $H_0 : \beta = 1$	0.0896(< 1:10)	0.0209(< 0.05)
Bayes factor B_{01}/p -Value: $H_0 : \kappa = \beta = 1$	0.0187(< 1:30)	0.0076(< 0.05)

The table shows the estimation results of aircraft generator data. The standard deviations of estimates are reported in square brackets which is reached through asymptotic normal distribution.

“Bayes 1997” denotes the vague prior proposed by Calabria and Pulcini [11] and the results of Bayes ($\pi_2(\mu)$) denotes the prior $\pi_2(\mu)$ proposed in our paper. For the Bayesian estimates, the iteration number of Gibbs sampling is also 5000 with the first 3000 samples as burn-in samples. The run time for the Bayesian estimates with $\pi_2(\mu)$ is 4.021 s and the run time for the maximum likelihood estimates (MLE) is 0.386 s.

The results with Black and Rigdon [10], Calabria and Pulcini [11] and our method reach the same conclusions: the system shows improvement (the upper limit on β is much less than 1) and the PLP is not an adequate model for these failure data (the 95% credibility interval on κ does not include the value 1).

Bayesian hypothesis testing is also applied to test whether a simpler model is suitable or not. We consider three hypotheses: $H_0 : \kappa = 1$ (NHPP), $H_0 : \beta = 1$ (Gamma RP) and $H_0 : \kappa = \beta = 1$ (HPP) versus $H_1 : \beta, \theta > 0, \kappa > 1$ (MPLP) is used, respectively. The priors of parameters in testing are also on the basis of $\pi_2(\mu)$ and the prior odds ratio between H_0 and H_1 is assumed to be 1. Bayes factors of H_0 to H_1 , say B_{01} , are calculated and listed in Table 8, in which we also calculate the p -Values based on the classical method. From Table 8, we can see that MPLP model is much more preferred.

7. Conclusion

In this paper, recurrent event data are modeled through modulated power law process (MPLP). Bayesian analyses are applied to this problem, where Jeffreys prior and six other reference priors are obtained. Interest is focused on four of them. Propriety of the corresponding posterior densities is proved and the predictive distributions of future failure time are derived additionally.

For the purpose of comparison, the simulation work and real data analysis are carried out based on both the objective Bayesian and maximum likelihood approaches, which show that the objective Bayesian estimation and prediction possess have much better statistical properties.

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Appendix

Theorem 4. For $\beta > 0$, $\theta > 0$ and $\kappa > 0$, the joint posterior distribution of μ is proper based on prior $\pi_{ab}(\mu)$ if sample size $n > a$.

Proof. We are going to prove the nominal constant $D < \infty$. First of all,

$$\begin{aligned} L(\mathbf{t}|\boldsymbol{\mu})\pi_{ab}(\boldsymbol{\mu}) &= \frac{\beta^{n-a}}{\Gamma(\kappa)^n \theta^{n\beta\kappa+1}} \prod_{i=1}^n t_i^{\beta-1} \left[\prod_{i=1}^n (t_i^\beta - t_{i-1}^\beta) \right]^{\kappa-1} \exp \left\{ -\left(\frac{t_n}{\theta}\right)^\beta \right\} [\psi^{(1)}(\kappa) - b/\kappa]^{1/2} \\ &\propto \frac{\beta^{n-a}}{\Gamma(\kappa)^n \theta^{n\beta\kappa+1}} \left(\prod_{i=1}^n t_i \right)^\beta \left[\prod_{i=1}^n (t_i^\beta - t_{i-1}^\beta) \right]^{\kappa-1} \exp \left\{ -\left(\frac{t_n}{\theta}\right)^\beta \right\} [\psi^{(1)}(\kappa) - b/\kappa]^{1/2}. \end{aligned}$$

Since $[\psi^{(1)}(\kappa) - b/\kappa]^{1/2} \leq [\psi^{(1)}(\kappa)]^{1/2}$ for $b = 0, 1$, it suffices to prove that

$$H_1 = \int_0^\infty \int_0^\infty \int_0^\infty \frac{\beta^{n-a}}{\Gamma(\kappa)^n \theta^{n\beta\kappa+1}} (\prod_{i=1}^n t_i)^\beta \left[\prod_{i=1}^n (t_i^\beta - t_{i-1}^\beta) \right]^{\kappa-1} \exp \left\{ -\left(\frac{t_n}{\theta}\right)^\beta \right\} [\psi^{(1)}(\kappa)]^{1/2} d\beta d\theta d\kappa < \infty.$$

Making the transformation $\xi = \theta^{-\beta}$,

$$\begin{aligned} H_1 &= \int_0^\infty \int_0^\infty \frac{\beta^{n-a-1}}{\Gamma(\kappa)^n} \left(\prod_{i=1}^n t_i \right)^\beta \left[\prod_{i=1}^n (t_i^\beta - t_{i-1}^\beta) \right]^{\kappa-1} [\psi^{(1)}(\kappa)]^{1/2} \left(\int_0^\infty \xi^{n\kappa-1} \exp\{-\xi t_n^\beta\} d\xi \right) d\beta d\kappa \\ &= \int_0^\infty \int_0^\infty \frac{\beta^{n-a-1} \Gamma(n\kappa)}{\Gamma(\kappa)^n t_n^{n\beta\kappa}} \left(\prod_{i=1}^n t_i \right)^\beta \left[\prod_{i=1}^n (t_i^\beta - t_{i-1}^\beta) \right]^{\kappa-1} [\psi^{(1)}(\kappa)]^{1/2} d\beta d\kappa \\ &= \int_0^\infty \int_1^\infty (\cdot) d\beta d\kappa + \int_0^\infty \int_0^1 (\cdot) d\beta d\kappa = H_2 + H_3, \end{aligned} \quad (23)$$

where $(\cdot)$ denotes the integrand from the previous line. In (23), H_1 is partitioned into two parts H_2 and H_3 according to the range of κ . We are therefore going to prove $H_2 < \infty$ and $H_3 < \infty$.

In [22], they proved the following sharp inequalities for $x > 1$:

$$\frac{x^{x-\gamma}}{e^{x-1}} < \Gamma(x) < \frac{x^{x-1/2}}{e^{x-1}},$$

where $\gamma = 0.577215\dots$ is the Euler–Mascheroni constant. Thus, for $\kappa > 1$,

$$\Gamma(n\kappa) < \frac{(n\kappa)^{n\kappa-1/2}}{e^{n\kappa-1}}, \quad \Gamma(\kappa) > \frac{\kappa^{\kappa-\gamma}}{e^{\kappa-1}} \quad \text{and} \quad \frac{\Gamma(n\kappa)}{\Gamma(\kappa)^n} < \kappa^{n\gamma-1/2} n^{n\kappa-1/2} e^{1-n}.$$

Moreover, $[\psi^{(1)}(\kappa)]^{1/2} < [\psi^{(1)}(1)]^{1/2}$ as $\psi^{(1)}(\kappa)$ is decreasing on $\kappa \in (1, \infty)$. Then we have

$$\begin{aligned} H_2 &= \int_0^\infty \int_1^\infty \frac{\beta^{n-a-1} \Gamma(n\kappa)}{\Gamma(\kappa)^n t_n^{\beta\kappa}} \left(\prod_{i=1}^n t_i \right)^\beta \left[\prod_{i=1}^n (t_i^\beta - t_{i-1}^\beta) \right]^{\kappa-1} [\psi^{(1)}(\kappa)]^{1/2} d\beta d\kappa \\ &< n^{n-1/2} e^{1-n} [\psi^{(1)}(1)]^{1/2} \int_0^\infty \int_1^\infty \beta^{n-a-1} \left(\prod_{i=1}^n \frac{t_i^\beta}{t_n^\beta} \right) \kappa^{n\gamma-1/2} \left(n^n \prod_{i=1}^n \frac{t_i^\beta - t_{i-1}^\beta}{t_n^\beta} \right)^{\kappa-1} d\beta d\kappa \\ &= n^{n-1/2} e^{1-n} [\psi^{(1)}(1)]^{1/2} \int_0^\infty \int_0^\infty \beta^{n-a-1} \left(\prod_{i=1}^n \frac{t_i^\beta}{t_n^\beta} \right) (\kappa+1)^{n\gamma-1/2} \left(n^n \prod_{i=1}^n \frac{t_i^\beta - t_{i-1}^\beta}{t_n^\beta} \right)^\kappa d\beta d\kappa \\ &< n^{n-1/2} e^{1-n} [\psi^{(1)}(1)]^{1/2} \int_0^\infty \int_0^\infty \beta^{n-a-1} \left(\prod_{i=1}^n \frac{t_i^\beta}{t_n^\beta} \right) (\kappa+1)^n \left(n^n \prod_{i=1}^n \frac{t_i^\beta - t_{i-1}^\beta}{t_n^\beta} \right)^\kappa d\beta d\kappa \\ &\propto \sum_{m=0}^n \binom{n}{m} \left\{ \int_0^\infty \int_0^\infty \beta^{n-a-1} \left(\prod_{i=1}^n \frac{t_i^\beta}{t_n^\beta} \right) \kappa^m \left(n^n \prod_{i=1}^n \frac{t_i^\beta - t_{i-1}^\beta}{t_n^\beta} \right)^\kappa d\beta d\kappa \right\} \\ &= \sum_{m=0}^n \binom{n}{m} \Gamma(m+1) \left\{ \int_0^\infty \beta^{n-a-1} \left(\prod_{i=1}^n \frac{t_i^\beta}{t_n^\beta} \right) \left[\log \left(\prod_{i=1}^n \frac{t_i^\beta}{n(t_i^\beta - t_{i-1}^\beta)} \right) \right]^{-m-1} d\beta \right\}, \end{aligned}$$

where the last “=” holds on the basis of the inequality of arithmetic and geometric means, i.e.

$$\prod_{i=1}^n \frac{n(t_i^\beta - t_{i-1}^\beta)}{t_n^\beta} = n^n \prod_{i=1}^n \frac{t_i^\beta - t_{i-1}^\beta}{t_n^\beta} < n^n \left(\frac{1}{n} \sum_{i=1}^n \frac{t_i^\beta - t_{i-1}^\beta}{t_n^\beta} \right)^n = 1.$$

Notice that for $m = 0, 1, \dots, n$,

$$\lim_{\beta \rightarrow 0} \left[\log \left(\prod_{i=1}^n \frac{t_i^\beta}{n(t_i^\beta - t_{i-1}^\beta)} \right) \right]^{-m-1} = \lim_{\beta \rightarrow \infty} \left[\log \left(\prod_{i=1}^n \frac{t_i^\beta}{n(t_i^\beta - t_{i-1}^\beta)} \right) \right]^{-m-1} = 0.$$

Then a constant $M > 0$ can be chosen such that

$$\left[\log \left(\prod_{i=1}^n \frac{t_i^\beta}{n(t_i^\beta - t_{i-1}^\beta)} \right) \right]^{-m-1} < M, \quad \text{for any } \beta > 0 \text{ and } m \in \{0, 1, \dots, n\}.$$

Thus when $n > a$,

$$\begin{aligned} &\sum_{m=0}^n \binom{n}{m} \Gamma(m+1) \left\{ \int_0^\infty \beta^{n-a-1} \left(\prod_{i=1}^n \frac{t_i^\beta}{t_n^\beta} \right) \left[\log \left(\prod_{i=1}^n \frac{t_i^\beta}{n(t_i^\beta - t_{i-1}^\beta)} \right) \right]^{-m-1} d\beta \right\} \\ &< \sum_{m=0}^n \binom{n}{m} \Gamma(m+1) M \int_0^\infty \beta^{n-a-1} \left(\prod_{i=1}^n \frac{t_i^\beta}{t_n^\beta} \right) d\beta \\ &= \sum_{m=0}^n \binom{n}{m} \Gamma(m+1) M \Gamma(n-a) \left\{ \log \left(\prod_{i=1}^n \frac{t_i^\beta}{t_n^\beta} \right) \right\}^{-n+a} < \infty \end{aligned}$$

Therefore, $H_2 < \infty$ holds.

On the other hand, note that

$$\lim_{\kappa \rightarrow 0} \frac{\Gamma(n\kappa)}{\Gamma(\kappa)^n} [\psi^{(1)}(\kappa)]^{1/2} = 0$$

and $\frac{\Gamma(n\kappa)}{\Gamma(\kappa)^n} [\psi^{(1)}(\kappa)]^{1/2}$ is continuous on $(0, 1]$. Thus $\frac{\Gamma(n\kappa)}{\Gamma(\kappa)^n} [\psi^{(1)}(\kappa)]^{1/2}$ can have an upper bound given $0 < \kappa \leq 1$. In

addition, $(\prod_{i=1}^n \frac{t_i^\beta - t_{i-1}^\beta}{t_n^\beta})^{\kappa-1}$ is also bounded given $\beta > 0$ and $0 < \kappa \leq 1$. Let U be such a constant satisfying

$$\frac{\Gamma(n\kappa)}{\Gamma(\kappa)^n} [\psi^{(1)}(\kappa)]^{1/2} \left(\prod_{i=1}^n \frac{t_i^\beta - t_{i-1}^\beta}{t_n^\beta} \right)^{\kappa-1} < U \quad \text{for any } \beta > 0, 0 < \kappa \leq 1.$$

Then we have, when $n > a$,

$$\begin{aligned}
H_3 &= \int_0^\infty \int_0^1 \frac{\beta^{n-a-1} \Gamma(n\kappa)}{\Gamma(\kappa)^n t_n^{\beta\kappa}} \left(\prod_{i=1}^n t_i \right)^\beta \left[\prod_{i=1}^n (t_i^\beta - t_{i-1}^\beta) \right]^{\kappa-1} [\psi^{(1)}(\kappa)]^{1/2} d\beta d\kappa \\
&< U \int_0^\infty \beta^{n-a-1} \left(\prod_{i=1}^n \frac{t_i^\beta}{t_n^\beta} \right) d\beta < \infty.
\end{aligned}$$

Based on the above inference,

$$H_1 = H_2 + H_3 < \infty, \quad \text{so that } D < \infty.$$

Consequently, if $n > a$, the posterior distribution considering reference prior $\pi_{ab}(\mu)$ is proper. $\square$

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